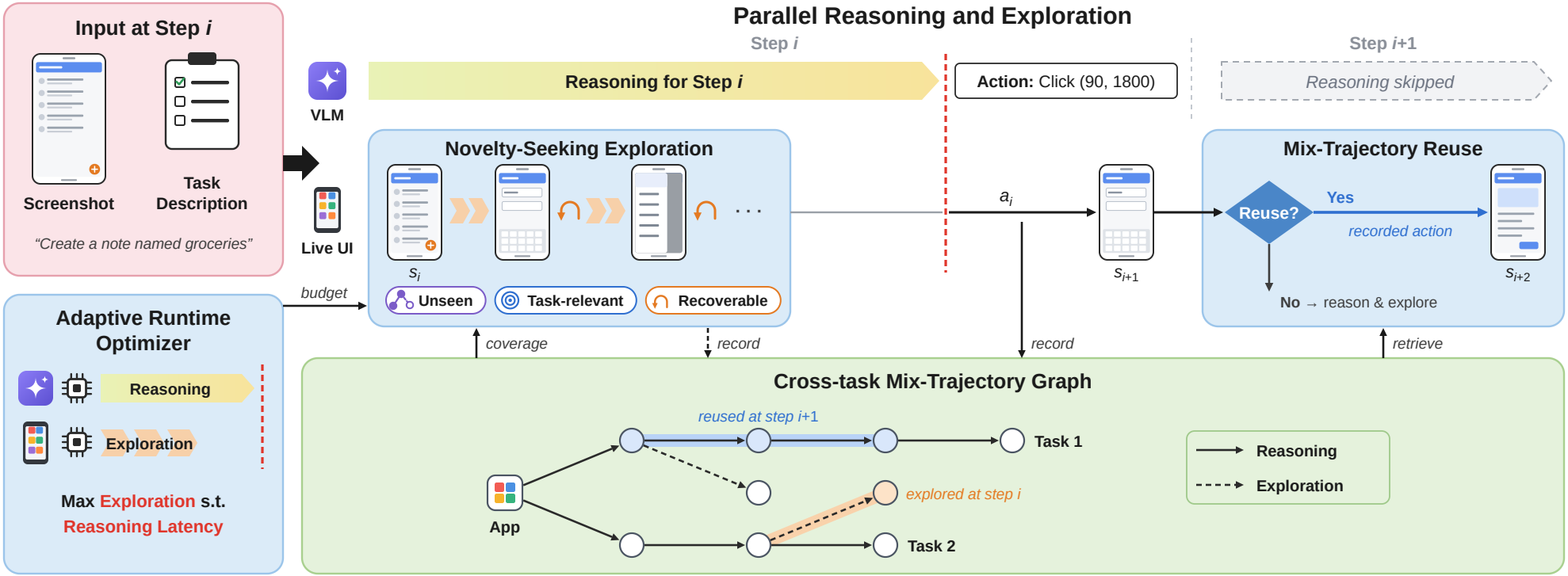

Parallel Reasoning and Exploration
Step i
Step i+1
Input at Step i
Screenshot
Task Description
"Create a note named groceries"
VLM
Live UI
Reasoning for Step i
Action: Click (90, 1800)
Reasoning skipped
Novelty-Seeking Exploration
s_i
Unseen
Task-relevant
Recoverable
a_i
s_{i+1}
Mix-Trajectory Reuse
Reuse?
Yes
recorded action
s_{i+2}
No → reason & explore
budget
Adaptive Runtime Optimizer
Reasoning
Exploration
Max Exploration s.t.
Reasoning Latency
coverage
record
record
retrieve
Cross-task Mix-Trajectory Graph
reused at step i+1
Task 1
App
explored at step i
Task 2
Reasoning
Exploration